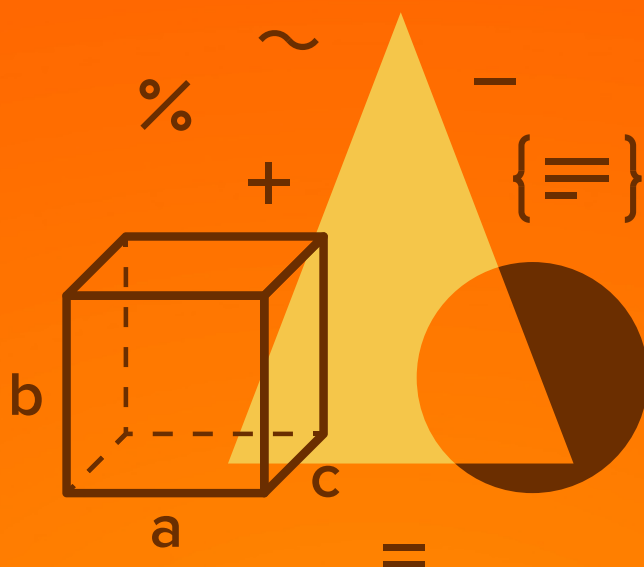




MATHEMATICS

JEE (MAIN+ADVANCED)

ENTHUSIAST COURSE

EXERCISE

Matrices

English Medium

EXERCISE (O-1)

Straight Objective Type

1. Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$, then $\text{Tr}(A) - \text{Tr}(B)$ has the value equal to

(A) 0 (B) 1 (C) 2 (D) none

MT0001

2. If $\begin{bmatrix} x & 3x-y \\ zx+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, then

(A) $x = 3, y = 7, z = 1, w = 14$ (B) $x = 3, y = -5, z = -1, w = -4$
 (C) $x = 3, y = 6, z = 2, w = 7$ (D) None of these

MT0002

3. The matrix $A^2 + 4A - 5I$, where I is an identity matrix and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$ equals :

(A) $32 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (B) $4 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$ (C) $4 \begin{bmatrix} 0 & -1 \\ 2 & 2 \end{bmatrix}$ (D) $32 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$

MT0003

4. A and B are two given matrices such that the order of A is 3×4 , if $A'B$ and BA' are both defined then

(A) order of B' is 3×4 (B) order of $B'A$ is 4×4
 (C) order of $B'A$ is 3×3 (D) $B'A$ is undefined

MT0005

5. If A, B and C are $n \times n$ matrices and $\det(A) = 2$, $\det(B) = 3$ and $\det(C) = 5$, then the value of the $\det(A^2BC^{-1})$ is equal to

(A) $\frac{6}{5}$ (B) $\frac{12}{5}$ (C) $\frac{18}{5}$ (D) $\frac{24}{5}$

MT0007

6. If $M = \begin{bmatrix} 0 & 2 \\ 5 & 0 \end{bmatrix}$ and $N = \begin{bmatrix} 0 & 5 \\ 2 & 0 \end{bmatrix}$, then M^{2011} is -

(A) $10^{1005}M$ (B) $10^{1005}N$ (C) $10^{2010}M$ (D) $10^{2011}M$

MT0008

7. Let three matrices be $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$, then

$$\text{tr}(A) + \text{tr}\left(\frac{ABC}{2}\right) + \text{tr}\left(\frac{A(BC)^2}{4}\right) + \text{tr}\left(\frac{A(BC)^3}{8}\right) + \dots + \infty \text{ is equal to-}$$

- (A) 6 (B) 9 (C) 12 (D) none

MT0009

8. For a matrix $A = \begin{bmatrix} 1 & 2r-1 \\ 0 & 1 \end{bmatrix}$, the value of $\prod_{r=1}^{50} \begin{bmatrix} 1 & 2r-1 \\ 0 & 1 \end{bmatrix}$ is equal to -

- (A) $\begin{bmatrix} 1 & 100 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 4950 \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 5050 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 2500 \\ 0 & 1 \end{bmatrix}$

MT0010

9. If the product of n matrices $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \dots \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is equal to the matrix $\begin{bmatrix} 1 & 378 \\ 0 & 1 \end{bmatrix}$ then the value of n is equal to -

- (A) 26 (B) 27 (C) 377 (D) 378

MT0011

10. Consider a matrix $A(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$ then

- (A) $A(\theta)$ is symmetric (B) $A(\theta)$ is skew symmetric
(C) $A^{-1}(\theta) = A(\pi - \theta)$ (D) $A^2(\theta) = A\left(\frac{\pi}{2} - 2\theta\right)$

MT0012

11. Which of the following is an orthogonal matrix -

- (A) $\begin{bmatrix} 6/7 & 2/7 & -3/7 \\ 2/7 & 3/7 & 6/7 \\ 3/7 & -6/7 & 2/7 \end{bmatrix}$ (B) $\begin{bmatrix} 6/7 & 2/7 & 3/7 \\ 2/7 & -3/7 & 6/7 \\ 3/7 & 6/7 & -2/7 \end{bmatrix}$
(C) $\begin{bmatrix} -6/7 & -2/7 & -3/7 \\ 2/7 & 3/7 & 6/7 \\ -3/7 & 6/7 & 2/7 \end{bmatrix}$ (D) $\begin{bmatrix} 6/7 & -2/7 & 3/7 \\ 2/7 & 2/7 & -3/7 \\ -6/7 & 2/7 & 3/7 \end{bmatrix}$

MT0013

12. The matrix $\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is

- (A) non-singular (B) Idempotent (C) Nilpotent (D) Involutory

MT0014

13. If $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$, then A is

- (A) Involutory matrix (B) Idempotent matrix (C) Nilpotent matrix (D) none of these

MT0015

14. Matrix $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, if $xyz = 60$ and $8x + 4y + 3z = 20$, then $A(\text{adj } A)$ is equal to -

(A) $\begin{bmatrix} 64 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 64 \end{bmatrix}$ (B) $\begin{bmatrix} 88 & 0 & 0 \\ 0 & 88 & 0 \\ 0 & 0 & 88 \end{bmatrix}$ (C) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$ (D) $\begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$

MT0017

15. Number of real values of λ for which the matrix $A = \begin{bmatrix} \lambda-1 & \lambda & \lambda+1 \\ 2 & -1 & 3 \\ \lambda+3 & \lambda-2 & \lambda+7 \end{bmatrix}$ has no inverse

- (A) 0 (B) 1 (C) 2 (D) infinite

MT0018

16. If $A = [a_{ij}]_{2 \times 2}$ where $a_{ij} = \begin{cases} i+j & i \neq j \\ i^2 - 2j & i = j \end{cases}$, then A^{-1} is equal to -

(A) $\frac{1}{9} \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$ (B) $\frac{1}{9} \begin{bmatrix} 0 & -3 \\ 3 & -1 \end{bmatrix}$ (C) $\frac{1}{9} \begin{bmatrix} 0 & -3 \\ -3 & -1 \end{bmatrix}$ (D) $\frac{1}{3} \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$

MT0020

17. If A and B are symmetric matrices and $AB = BA$, then $A^{-1}B$ is a -

- (A) symmetric matrix (B) skew symmetric matrix
(C) unit matrix (D) Nilpotent matrix

MT0127

18. If $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $\det. (A^n - I) = 1 - \lambda^n$, $n \in \mathbb{N}$ then the value of λ , is - **MT0128**
- (A) 1 (B) 2 (C) 3 (D) 4
19. If an idempotent matrix is also skew symmetric then it must be- **MT0129**
- (A) an involutory matrix (B) an identity matrix
(C) an orthogonal matrix (D) a null matrix
20. Let $A = \begin{bmatrix} a & 1 \\ -1 & b \end{bmatrix}$ where a and b are real number. If A^2 is a null matrix then the product ab equals- **MT0130**
- (A) 0 (B) 1 (C) -1 (D) ± 1
21. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & c \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$, then - **MT0131**
- (A) $a = 1, c = -1$ (B) $a = 2, c = -\frac{1}{2}$ (C) $a = -1, c = 1$ (D) $a = \frac{1}{2}, c = \frac{1}{2}$
22. If A is skew symmetric matrix of order 3 and X be another matrix of same order, then $|XA + AX^T|$ is (where $|P|$ denotes determinant of matrix P) - **MT0132**
- (A) $|X + X^T|$ (B) $|A + X|$ (C) $|A - X|$ (D) 0
23. If A and B are symmetric matrices of the same order and $X = AB + BA$, $Y = AB - BA$ then $(XY)^T$ is equal to : **MT0133**
- (A) XY (B) YX (C) $-YX$ (D) None
24. Let $A = \begin{bmatrix} \ell-3 & a & b \\ c & 6 & d \\ e & f & 9-\ell \end{bmatrix}$, $B = \text{adj}(A)$ and $C = \text{adj}(B)$. If $|A| = 5$, then $\text{tr}(C)$ is (where $|X|$, $\text{tr}(X)$ & $\text{adj}(X)$ denote determinant value, trace and adjoint of matrix X respectively) - **MT0134**
- (A) 5 (B) 12 (C) 30 (D) 60
25. $AB = A$ and $BA = B$, then (here A & B are matrix of $n \times n$) which of the following must be true - **MT0135**
- (A) $A = B$ (B) $A^2 = A$ (C) $A = I$ (D) $B = I$
26. The number of 3×3 non-singular matrices, with four entries as 1 and all other entries as 0, is :- **MT0077**
- (A) Less than 4 (B) 5 (C) 6 (D) At least 7

27. Let A be a 2×2 matrix with non-zero entries and let $A^2 = I$, where I is 2×2 identity matrix. Define $\text{Tr}(A)$ = sum of diagonal elements of A and $|A|$ = determinant of matrix A .

Statement-1 : $\text{Tr}(A) = 0$.

Statement-2 : $|A| = 1$.

- (A) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for statement-1.
 (C) Statement-1 is true, Statement-2 is false.
 (D) Statement-1 is false, Statement-2 is true.

MT0078

28. Let A and B be two symmetric matrices of order 3.

Statement-1 : $A(BA)$ and $(AB)A$ are symmetric matrices.

Statement-2 : AB is symmetric matrix if matrix multiplication of A with B is commutative.

- (A) Statement-1 is true, Statement-2 is false.
 (B) Statement-1 is false, Statement-2 is true
 (C) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1
 (D) Statement-1 is true, Statement-2 is true; Statement-2 is **not** a correct explanation for Statement-1.

MT0079

29. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. If u_1 and u_2 are column matrices such that $Au_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $Au_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, then

$u_1 + u_2$ is equal to :

- (A) $\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$ (B) $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ (C) $\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ (D) $\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$

MT0081

30. If P is a 3×3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the 3×3 identity matrix,

then there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that

- (A) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (B) $PX = X$ (C) $PX = 2X$ (D) $PX = -X$

MT0105

EXERCISE (O-2)

Multiple Correct Answer Type

1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & -1 \\ 3 & 0 & k \end{bmatrix}$ and $f(x) = x^3 - 2x^2 - \alpha x + \beta = 0$. If A satisfies $f(x) = 0$, then-

(A) $k = 1, \alpha = 14$ (B) $\alpha = 14, \beta = 22$ (C) $k = -1, \beta = 22$ (D) $\alpha = -14, \beta = -22$

MT0029

2. If D_1 and D_2 are two 3×3 diagonal matrices where none of the diagonal element is zero, then -

(A) $D_1 D_2$ is a diagonal matrix (B) $D_1 D_2 = D_2 D_1$
(C) $D_1^2 + D_2^2$ is a diagonal matrix (D) $D_1^2 + D_2^2 = (D_1 + D_2)^2$

MT0031

3. If A & B are two non singular matrices of order 3×3 such that $A^T + B = I$ & $BA^T = -B$, then which is/are always true (where X^T denotes transpose of X and I denotes unit matrix)-

(A) $|B| = 2$ (B) $|B| = 8$ (C) $|A| = -1$ (D) $|A| = 1$

(where $|X|$ denotes determinant value of X)

MT0037

4. If A and B are two 3×3 matrices such that their product AB is a null matrix then

(A) $\det. A \neq 0 \Rightarrow B$ must be a null matrix.
(B) $\det. B \neq 0 \Rightarrow A$ must be a null matrix.
(C) If none of A and B are null matrices then atleast one of the two matrices must be singular.
(D) If neither $\det. A$ nor $\det. B$ is zero then the given statement is not possible.

MT0032

5. If A and B are two orthogonal matrices of order 3, then -

(A) A and B both will be invertible matrices (B) matrix ABA will also be orthogonal
(C) matrix $A^2 B^2$ will also be orthogonal (D) maximum value of $\det\left(\frac{A}{2} \text{adj}(2B)\right)$ is 8.

MT0035

6. Let $\det(\text{adj}(\text{adj}A)) = 14^4$ where $A = \begin{bmatrix} x & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$, $x \neq -\frac{25}{3}$, then

(A) $x = 1$ (B) $\det(2A) = 112$ (C) $x = 2$ (D) $\det(2A) = 256$

MT0038

7. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$ & $A^n = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, (where $n \geq 2$ & $n \in \mathbb{N}$), then -

(A) $a = d$

(B) $b = c$

(C) $b = a + 1$ if n is odd

(D) $b = a - 1$ if n is even

MT0036

8. If A & B are square matrices of order 2 such that $A + \text{adj}(B^T) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ & $A^T - \text{adj}(B) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, then-

(A) B is symmetric matrix

(B) $A^n = A \forall n \in \mathbb{N}$

(C) $|A + A^2 + A^3 + A^4 + A^5| = 0$

(D) $|B + B^2 + B^3 + B^4 + B^5| = 0$

MT0034

9. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P

is (are) -

(A) -2

(B) -1

(C) 1

(D) 2

MT0106

10. Consider the following statements.

Statement-1 : If $\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} y & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$, then $2x + 3y = \lambda$.

Statement-2 : Given that $\ell + 5 = p + 2m$, where

A is a square matrix of order n .

ℓ = maximum number of distinct entries if A is a triangular matrix

m = maximum number of distinct entries if A is a diagonal matrix

p = minimum number of zeroes if A is a triangular matrix.

Statement-3 : Let A be the set of all 3×3 skew symmetric matrices whose entries are either -1 , 0 or 1 . If there are exactly three 0 's, three 1 's and three (-1) 's, then number of such matrices is equal to μ .

Then, which of the following options is/are correct ?

(A) $2\lambda + 3n = 30$

(B) $3\lambda - 3\mu = 5$

(C) $4\mu - 2n = 24$

(D) $5n - 3\lambda + 2\mu = 10$

MT0150

11. Consider the following statements

Statement-1 : If A is an idempotent non-zero matrix and I is an identity matrix of the same order, such that $(A + I)^n = I + 127A$, ($n \in \mathbb{N}$), then 'n' has 3 positive divisors.

Statement-2 : Let $A = \begin{bmatrix} 3x^2 & & \\ & 1 & \\ & & 6x \end{bmatrix}$, $B = [a \ b \ c]$ and $C = \begin{bmatrix} (x+2)^2 & 5x^2 & 2x \\ 5x^2 & 2x & (x+2)^2 \\ 2x & (x+2)^2 & 5x^2 \end{bmatrix}$ be three given

matrices, where a, b, c and $x \in \mathbb{R}$. Given that $\text{tr}(AB) = \text{tr}(C) \forall x \in \mathbb{R}$, where $\text{tr}(A)$ denotes trace of A . Solving, we get $a + b + c = 7$.

Then, which of the following options is/are correct ?

(A) Statement-1 is True

(B) Statement-1 is False

(C) Statement-2 is True

(D) Statement-2 is False

MT0151

12. Which of the following options is/are correct ?

(A) Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$ are two matrices such that $AB = (AB)^{-1}$ and $AB \neq I$

(where I is an identity matrix of order 3×3). Then $\text{tr}(AB + (AB)^2 + (AB)^3 + \dots + (AB)^{100})$ equals 100.

(B) If A and B are square matrices of order 3, where $|A| = -2$ and $|B| = 1$, then $|(A^{-1})\text{adj}(B^{-1})\text{adj}(2A^{-1})|$ equals 8.

(C) If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then $F(x) \cdot F(y) = F(x+y)$ and $[F(x)]^{-1} = F(-x)$.

(D) Let X be the solution set of the equation $A^x = I$, where $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ and I is the corresponding

unit matrix and $X \subseteq \mathbb{N}$ then the minimum value of $\sum (\cos^x \theta + \sin^x \theta), \theta \in \mathbb{R} - \left\{ \frac{n\pi}{2}, n \in \mathbb{Z} \right\}$ is 2.

MT0152

13. Consider the following statements

Statement-1 : Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$. If $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, then λ denotes sum of elements of P.

Statement-2 : Let μ denote the sum of elements of the matrix A satisfying the matrix equation,

$$\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \cdot A \cdot \begin{bmatrix} 3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & -1 \end{bmatrix}$$

Statement-3 : Given that $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 10 \\ 13 \\ 9 \end{bmatrix}$ and that $Cb = D$. If $AX = b$, then

v denotes the sum of elements of X.

Then, which of the following options is/are correct ?

(A) $2\lambda - 19\mu = 3$

(B) $38\mu + 15v = 3$

(C) $10v + 8\mu = 2$

(D) $\lambda + 19\mu + 38\mu = 0$

MT0153

14. Which of the following options is/are correct ?

(A) Let $A = \begin{bmatrix} 1 & \frac{3}{2} \\ 2 & 2 \\ 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 4 & -3 \\ -2 & 2 \end{bmatrix}$ and $C_r = \begin{bmatrix} r \cdot 3^r & 2^r \\ 0 & (r-1)3^r \end{bmatrix}$ be 3 given matrices. Then

$$\sum_{r=1}^{50} \text{tr}((AB)^r C_r) = 3(49 \cdot 3^{50} + 1).$$

(B) Given $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$. I is a unit matrix of order 2. If $AX = A$, then $X = \begin{bmatrix} a & b \\ 2-2a & 1-2b \end{bmatrix}$ for

$$a, b \in \mathbb{R}$$

(C) Given $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$. I is a unit matrix of order 2. If $XA = I$, then X does not exist

(D) Given $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$. I is a unit matrix of order 2. If $XB = \mathbf{O}$ but $BX \neq \mathbf{O}$, then

$$X = \begin{bmatrix} a & -3a \\ c & -3c \end{bmatrix} \quad a, c \in \mathbb{R}, 3a + c \neq 0; 3b + d \neq 0$$

MT0154

15. If A is a non-singular matrix and A^T denotes the transpose of A, then :
- (A) $|A| \neq |A^T|$ (B) $|A \cdot A^T| = |A|^2$
(C) $|A^T \cdot A| = |A^T|^2$ (D) $|A| + |A^T| \neq 0$ **MT0155**
16. Let A, B, C, D be real matrices such that $A^T = BCD$; $B^T = CDA$; $C^T = DAB$ and $D^T = ABC$ for the matrix $M = ABCD$, then M^{2016} is equal to
- (A) M (B) M^2 (C) M^3 (D) M^4 **MT0156**
17. If $A^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix}$, then
- (A) $|A| = 2$ (B) A is non-singular
(C) $\text{Adj.} A = \begin{bmatrix} 1/2 & -1/2 & 0 \\ 0 & -1 & 1/2 \\ 0 & 0 & -1/2 \end{bmatrix}$ (D) A is skew symmetric matrix **MT0157**
18. If A and B are square matrices of order 3, then the true statement is/are (where I is unit matrix).
- (A) $\det(-A) = -\det A$
(B) If AB is singular then atleast one of A or B is singular
(C) $\det(A + I) = 1 + \det A$
(D) $\det(2A) = 2^3 \det A$ **MT0158**
19. Let M be a 3×3 non-singular matrix with $\det(M) = 4$. If $M^{-1} \text{adj}(\text{adj } M) = k^2 I$, then the value of 'k' may be :
- (A) +2 (B) 4 (C) -2 (D) -4 **MT0159**
20. If $AX = B$ where A is 3×3 and X and B are 3×1 matrices then which of the following is correct ?
- (A) If $|A| = 0$ then $AX = B$ has infinite solutions
(B) If $AX = B$ has infinite solutions then $|A| = 0$
(C) If $(\text{adj}(A))B = 0$ and $|A| \neq 0$ then $AX = B$ has unique solution
(D) If $(\text{adj}(A))B \neq 0$ & $|A| = 0$ then $AX = B$ has no solution **MT0160**

EXERCISE (O-3)**Linked Comprehension Type****Paragraph for Question 1 and 2**

Consider the system $AX = B$, where $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ 3 & 2 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$.

1. Sum of elements of $(\text{adj}A) B$ is -
 (A) -1 (B) 2 (C) -2 (D) -4

MT0041

2. Value of $\text{tr}(XB^T)$ is (where $\text{tr}(A)$ denotes trace of matrix A) -
 (A) 0 (B) 1 (C) 2 (D) 3

MT0041

Paragraph for Question 3 to 5

If A is a symmetric and B skew symmetric matrix and $A + B$ is non singular and $C = (A + B)^{-1}(A - B)$ then

3. $C^T(A + B)C =$
 (A) $A + B$ (B) $A - B$ (C) A (D) B

MT0042

4. $C^T(A - B)C =$
 (A) $A + B$ (B) $A - B$ (C) A (D) B

MT0042

5. $C^TAC =$
 (A) $A + B$ (B) $A - B$ (C) A (D) B

MT0042

Paragraph for Question 6 to 8

Let p be an odd prime number and T_p be the following set of 2×2 matrices :

$$T_p = \left\{ A = \begin{bmatrix} a & b \\ c & a \end{bmatrix} : a, b, c \in \{0, 1, 2, \dots, p-1\} \right\}$$

6. The number of A in T_p such that A is either symmetric or skew-symmetric or both, and $\det(A)$ divisible by p is -
 (A) $(p-1)^2$ (B) $2(p-1)$ (C) $(p-1)^2 + 1$ (D) $2p-1$
7. The number of A in T_p such that the trace of A is not divisible by p but $\det(A)$ is divisible by p is -
 [Note : The trace of a matrix is the sum of its diagonal entries.]
 (A) $(p-1)(p^2 - p + 1)$ (B) $p^3 - (p-1)^2$
 (C) $(p-1)^2$ (D) $(p-1)(p^2 - 2)$
8. The number of A in T_p such that $\det(A)$ is not divisible by p is -
 (A) $2p^2$ (B) $p^3 - 5p$ (C) $p^3 - 3p$ (D) $p^3 - p^2$

MT0100

Matrix Match Type

9. Let $\mathcal{A}$ be the set of all 3×3 symmetric matrices all of whose entries are either 0 or 1. Five of these entries are 1 and four of them are 0.

List-I

- (A) The number of matrices in $\mathcal{A}$ is
(B) The number of matrices A in $\mathcal{A}$ for which the system

of linear equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has a unique solution, is

- (C) The number of matrices A in $\mathcal{A}$ for which the system of

linear equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ is inconsistent, is

- (D) The number of matrices A in $\mathcal{A}$ for which the system of

linear equations $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has infinitely many solutions, is

List-II

- (P) more than 2
(Q) 12

- (R) more than 4

- (S) less than 7

- (A) I-Q, II-P,R,S, III-P,S, IV-S
(C) I-P,R, II-P,S, III-Q, IV-R,S

- (B) I-Q, II-Q,S, III-P, IV-S
(D) I-R,S, II-Q,S, III-P, IV-P,Q

MT0161

10. **Column-I**

- (A) Let $\omega \neq 1$ be a cube root of unity and S be the set of

all non-singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$,

where each of a, b and c is either ω or ω^2 . Then the number of distinct matrices in the set S is-

- (B) Let M be 3×3 matrix satisfying $M \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$, $M \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

- (Q) 4

and $M \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$. Then the sum of the diagonal entries of M is

- (C) The number of 3×3 matrices A whose entries are either 0 or 1 and (R) 9

for which the system $A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has exactly two distinct solutions, is

- (D) Let k be a positive real number and let (S) 2

$A = \begin{bmatrix} 2k-1 & 2\sqrt{k} & 2\sqrt{k} \\ 2\sqrt{k} & 1 & -2k \\ -2\sqrt{k} & 2k & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2k-1 & \sqrt{k} \\ 1-2k & 0 & 2\sqrt{k} \\ -\sqrt{k} & -2\sqrt{k} & 0 \end{bmatrix}$.

If $\det(\text{adj } A) + \det(\text{adj } B) = 10^6$, then $[k]$ is equal to
[Note : $\text{adj } M$ denotes the adjoint of a square matrix M and $[k]$ denotes the largest integer less than or equal to k].

MT0162

EXERCISE (O-4)

Numerical Grid Type

1. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $A^2 - kA - I_2 = 0$, then value of k is - MT0004
2. If p, q, r are 3 real number satisfying the matrix equation, $\begin{bmatrix} p & q & r \end{bmatrix} \begin{bmatrix} 3 & 4 & 1 \\ 3 & 2 & 3 \\ 2 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 1 \end{bmatrix}$, then $2p + q - r$ equals :- MT0006
3. Find the number of $A = [a_{ij}]_{2 \times 2}$ satisfying a_{ij} is 1 or -1 and $a_{11}a_{21} + a_{12}a_{22} = 0$. MT0043
4. Let $M = \begin{bmatrix} a & -360 \\ b & c \end{bmatrix}$, where a, b and c are integers. Find the smallest positive value of b such that $M^2 = \mathbf{O}$, where $\mathbf{O}$ denotes 2×2 null matrix. MT0045
5. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ are matrices satisfying $AB = B$ and $a + d = 5050$. Find the value of $(ad - bc)$. MT0046
6. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2, then the number of positive divisors of determinant of the matrix Q is - MT0104
7. Let A, B and $A + B$ are non-singular matrices of order 3×3 satisfying $A^{-1} + B^{-1} = (A + B)^{-1}$ and $|AB^{-1}| \in \mathbb{R}$ then value of $\frac{|A|}{|B|}$ is MT0138
8. Let P be a 2×2 matrix such that $P \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $P^2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. If p_1 and p_2 ($p_1 > p_2$) are two values of p for which $\det(P - pI) = 0$, where I is an identity matrix of order 2, then $(5p_1 + 2p_2)$ is equal to MT0139
[Note : $\det(M)$ denotes determinant of square matrix M]
9. Number of 3×3 symmetric matrices which can be formed by three '0', three '1' & three '-1' only, is MT0140
10. An invertible matrix A of order 3 satisfies the relation $A = A^{-1} + 2I$, (where I denotes identity matrix). The value of $|A - I| \cdot |A + I| \cdot |A - 2I|$ is MT0141

EXERCISE (JM)

1. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to

[JEE(Main) - 2013]

- (1) 4 (2) 11 (3) 5 (4) 0

MT0082

2. If A is an 3×3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1} A'$, the BB' equals :

[JEE(Main) - 2014]

- (1) $I + B$ (2) I (3) B^{-1} (4) $(B^{-1})'$

MT0083

3. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the

ordered pair (a, b) is equal to :

[JEE(Main)-2015]

- (1) $(2, 1)$ (2) $(-2, -1)$ (3) $(2, -1)$ (4) $(-2, 1)$

MT0084

4. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \text{ adj } A = AA^T$, then $5a + b$ is equal to :

[JEE(Main)-2016]

- (1) 13 (2) -1 (3) 5 (4) 4

MT0085

5. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then $\text{adj } (3A^2 + 12A)$ is equal to :-

[JEE(Main)-2017]

- (1) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$ (2) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$ (3) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$ (4) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$

MT0086

6. Let A and B be two invertible matrices of order 3×3 . If $\det(ABA^T) = 8$ and $\det(AB^{-1}) = 8$, then $\det(BA^{-1} B^T)$ is equal to :-

[JEE(Main) -2019]

- (1) 16 (2) $\frac{1}{16}$ (3) $\frac{1}{4}$ (4) 1

MT0087

7. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$. Then $\frac{q_{21} + q_{31}}{q_{32}}$ is equal to :

[JEE(Main) -2019]

- (1) 15 (2) 9 (3) 135 (4) 10

MT0088

8. Let $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, ($\alpha \in \mathbb{R}$) such that $A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then a value of α is

[JEE(Main) -2019]

- (1) $\frac{\pi}{16}$ (2) 0 (3) $\frac{\pi}{32}$ (4) $\frac{\pi}{64}$

MT0089

9. If $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$, then the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is

[JEE(Main) -2019]

- (1) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$ (3) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$

MT0090

10. If A is a symmetric matrix and B is a skew-symmetric matrix such that $A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to :

[JEE(Main) -2019]

- (1) $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$ (2) $\begin{bmatrix} -4 & -2 \\ -1 & 4 \end{bmatrix}$ (3) $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$ (4) $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$

MT0091

11. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)}a_{ji}$, where $i, j = 1, 2, 3$. If the determinant of B is 81, then the determinant of A is :

[JEE(Main) -2020]

- (1) 3 (2) $1/3$ (3) $1/81$ (4) $1/9$

MT0092

12. Let α be a root of the equation $x^2 + x + 1 = 0$ and the matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$, then the matrix A^{31}

is equal to :

[JEE(Main) -2020]

- (1) A^3 (2) A (3) A^2 (4) I_3

MT0093

13. If $A = \begin{pmatrix} 2 & 2 \\ 9 & 4 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, then $10A^{-1}$ is equal to [JEE(Main) -2020]
 (1) $4I - A$ (2) $A - 6I$ (3) $6I - A$ (4) $A - 4I$

MT0094

14. The number of all 3×3 matrices A , with entries from the set $\{-1, 0, 1\}$ such that the sum of the diagonal elements of AA^T is 3, is [JEE(Main) -2020]

MT0095

15. If the matrices $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, $B = \text{adj}A$ and $C = 3A$, then $\frac{|\text{adj}B|}{|C|}$ is equal to :

[JEE(Main) -2020]

- (1) 72 (2) 2 (3) 8 (4) 16

MT0096

16. If the matrix $A = \begin{pmatrix} 0 & 2 \\ K & -1 \end{pmatrix}$ satisfies $A(A^3 + 3I) = 2I$, then the value of K is :

[JEE(Main) -2021]

- (1) $\frac{1}{2}$ (2) $-\frac{1}{2}$ (3) -1 (4) 1

MT0163

17. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some

non-zero $k \in \mathbb{R}$. If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to _____.

[JEE(Main) -2021]

MT0164

18. Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} ; a, b \in \{1, 2, 3, \dots, 100\} \right\}$ and let $T_n = \{A \in S : A^{n(n+1)} = I\}$. Then the number of elements in $\bigcap_{n=1}^{100} T_n$ is _____.

[JEE(Main) -2022]

MT0165

19. Let $A = \begin{bmatrix} 0 & -2 \\ 2 & 0 \end{bmatrix}$. If M and N are two matrices given by $M = \sum_{k=1}^{10} A^{2k}$ and $N = \sum_{k=1}^{10} A^{2k-1}$ then MN^2 is

[JEE(Main) -2022]

- (A) a non-identity symmetric matrix
 (B) a skew-symmetric matrix
 (C) neither symmetric nor skew-symmetric matrix
 (D) an identity matrix

MT0166

EXERCISE (JA)

1. For 3×3 matrices M and N , which of the following statement(s) is (are) **NOT** correct ?

- (A) $N^T M N$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric
(B) $MN - NM$ is skew symmetric for all symmetric matrices M and N
(C) MN is symmetric for all symmetric matrices M and N
(D) $(\text{adj } M)(\text{adj } N) = \text{adj } (M N)$ for all invertible matrices M and N

[JEE-Advanced 2013, 4, (-1)]

MT0107

2. Let M be a 2×2 symmetric matrix with integer entries. Then M is invertible if

- (A) the first column of M is the transpose of the second row of M
(B) the second row of M is the transpose of the first column of M
(C) M is a diagonal matrix with nonzero entries in the main diagonal
(D) the product of entries in the main diagonal of M is not the square of an integer

[JEE(Advanced)-2014, 3]

MT0108

3. Let M and N be two 3×3 matrices such that $MN = NM$. Further, if $M \neq N^2$ and $M^2 = N^4$, then

- (A) determinant of $(M^2 + MN^2)$ is 0
(B) there is a 3×3 non-zero matrix U such that $(M^2 + MN^2)U$ is zero matrix
(C) determinant of $(M^2 + MN^2) \geq 1$
(D) for a 3×3 matrix U , if $(M^2 + MN^2)U$ equals the zero matrix then U is the zero matrix

[JEE(Advanced)-2014, 3]

MT0109

4. Let X and Y be two arbitrary, 3×3 , non-zero, skew-symmetric matrices and Z be an arbitrary 3×3 , non-zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric ?

[JEE(Advanced)-2015, 4M, -2M]

- (A) $Y^3 Z^4 - Z^4 Y^3$ (B) $X^{44} + Y^{44}$ (C) $X^4 Z^3 - Z^3 X^4$ (D) $X^{23} + Y^{23}$

MT0110

5. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$, Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = kI$, where $k \in \mathbb{R}$,

$k \neq 0$ and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then -

- (A) $\alpha = 0, k = 8$ (B) $4\alpha - k + 8 = 0$
(C) $\det(\text{Padj}(Q)) = 2^9$ (D) $\det(\text{Qadj}(P)) = 2^{13}$

[JEE(Advanced)-2016, 4(-2)]

MT0111

6. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that

$P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}}$ equals

[JEE(Advanced)-2016, 3(-1)]

- (A) 52 (B) 103 (C) 201 (D) 205

MT0112

7. Which of the following is(are) NOT the square of a 3×3 matrix with real entries ?

[JEE(Advanced)-2017, 4(-2)]

- (A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

MT0113

8. How many 3×3 matrices M with entries from $\{0,1,2\}$ are there, for which the sum of the diagonal entries of $M^T M$ is 5 ?

[JEE(Advanced)-2017, 3(-1)]

- (A) 198 (B) 126 (C) 135 (D) 162

MT0114

9. For a real number α , if the system

$$\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

of linear equations, has infinitely many solutions, then $1 + \alpha + \alpha^2 =$

[JEE(Advanced)-2017, 3]

MT0115

10. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_3 \in \mathbb{R}$ and the system of equations (in real variables)

$$-x + 2y + 5z = b_1$$

$$2x - 4y + 3z = b_2$$

$$x - 2y + 2z = b_3$$

has at least one solution. Then, which of the following system(s) (in real variables) has (have) at

least one solution of each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$?

[JEE(Advanced)-2018, 4(-2)]

- (A) $x + 2y + 3z = b_1$, $4y + 5z = b_2$ and $x + 2y + 6z = b_3$
 (B) $x + y + 3z = b_1$, $5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$
 (C) $-x + 2y - 5z = b_1$, $2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$
 (D) $x + 2y + 5z = b_1$, $2x + 3z = b_2$ and $x + 4y - 5z = b_3$

MT0116

11. Let P be a matrix of order 3×3 such that all the entries in P are from the set $\{-1, 0, 1\}$. Then, the maximum possible value of the determinant of P is _____

[JEE(Advanced)-2018, 3(0)]

MT0117

12. Let $M = \begin{bmatrix} \sin^4 \theta & -1 - \sin^2 \theta \\ 1 + \cos^2 \theta & \cos^4 \theta \end{bmatrix} = \alpha I + \beta M^{-1}$,

where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real number, and I is the 2×2 identity matrix. If

α^* is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and

β^* is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi)\}$,

then the value of $\alpha^* + \beta^*$ is

[JEE(Advanced)-2019]

(1) $-\frac{37}{16}$

(2) $-\frac{29}{16}$

(3) $-\frac{31}{16}$

(4) $-\frac{17}{16}$

MT0142

13. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and $\text{adj}M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$ where a and b are real numbers. Which of the following options is/are correct ?

[JEE(Advanced)-2019]

(1) $a + b = 3$

(2) $\det(\text{adj}M^2) = 81$

(3) $(\text{adj}M)^{-1} + \text{adj}M^{-1} = -M$

(4) If $M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\alpha - \beta + \gamma = 3$ MT0143

14. Let $P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$,
 $P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ and $X = \sum_{k=1}^6 P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$ [JEE(Advanced)-2019]

where P_k^T denotes the transpose of the matrix P_k . Then which of the following options is/are correct?

- (1) $X - 30I$ is an invertible matrix (2) The sum of diagonal entries of X is 18

- (3) If $X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\alpha = 30$ (4) X is a symmetric matrix MT0144

15. Let $x \in \mathbb{R}$ and let $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$, $Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix}$ and $R = PQP^{-1}$.

Then which of the following options is/are correct ?

[JEE(Advanced)-2019]

- (1) For $x = 1$, there exists a unit vector $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ for which $R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

- (2) There exists a real number x such that $PQ = QP$

- (3) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$, for all $x \in \mathbb{R}$

- (4) For $x = 0$, if $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$, then $a + b = 5$ MT0145

16. Let M be a 3×3 invertible matrix with real entries and let I denote the 3×3 identity matrix. If $M^{-1} = \text{adj}(\text{adj } M)$, then which of the following statement is/are ALWAYS TRUE ?

[JEE(Advanced)-2020]

- (A) $M = I$ (B) $\det M = 1$ (C) $M^2 = I$ (D) $(\text{adj } M)^2 = I$

MT0146

17. The trace of a square matrix is defined to be the sum of its diagonal entries. If A is a 2×2 matrix such that the trace of A is 3 and the trace of A^3 is -18 , then the value of the determinant of A is

[JEE(Advanced)-2020]

MT0147

18. For any 3×3 matrix M , let $|M|$ denote the determinant of M . Let

[JEE(Advanced)-2021]

$$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$$

If Q is a nonsingular matrix of order 3×3 , then which of the following statements is (are) **TRUE**?

(A) $F = PEP$ and $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $|EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$

(C) $|(EF)^3| > |EF|^2$

(D) Sum of the diagonal entries of $P^{-1}EP + F$ is equal to the sum of diagonal entries of $E + P^{-1}FP$

MT0148

19. For any 3×3 matrix M , let $|M|$ denote the determinant of M . Let I be the 3×3 identity matrix. Let E and F be two 3×3 matrices such that $(I - EF)$ is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) **TRUE** ?

[JEE(Advanced)-2021]

(A) $|FE| = |I - FE||FGE|$

(B) $(I - FE)(I + FGE) = I$

(C) $EFG = GEF$

(D) $(I - FE)(I - FGE) = I$

MT0149

20. Let β be a real number. Consider the matrix

$$A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$$

If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of 9β is _____.

[JEE(Advanced)-2022]

MT0167

ANSWER KEY

Do yourself -1

1. $\begin{bmatrix} 3 & 5 & 7 \\ 4 & 6 & 8 \end{bmatrix}$
2. 6
3. 30
4. $x = 2, y = -1, a = 1, b = -3$
5. 8
6. $\begin{bmatrix} -7 & -10 & -7 \\ -2 & 6 & 3 \end{bmatrix}$ & 2×3
7. $\begin{bmatrix} -8 & -9 \\ -7/2 & -8 \\ -1 & 1 \end{bmatrix}$

Do yourself - 2

3. $x = -2$
4. 5
5. 0

Do yourself -3

2. C
3. A
4. C
5. B
6. C
7. C
8. 6

Do yourself - 4

1. C
2. $x = 0$
3. A
4. D
5. A
6. A,C
7. $f(a) = 1/4, a = 1/2$

Do yourself - 5

2. $\begin{bmatrix} 1 & 6 & -9 \\ 1 & 6 & -3 \end{bmatrix}$
5. C
6. C
7. 1
8. 650

Do yourself - 6

1. -4
2. $\begin{bmatrix} 2 & 7 & 4 \\ 7 & -7 & \frac{1}{2} \\ 4 & \frac{1}{2} & 0 \end{bmatrix} + \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & \frac{3}{2} \\ -3 & -\frac{3}{2} & 0 \end{bmatrix}$
3. B
4. A,B,C
5. B
6. A
7. D
8. 6
9. A
10. A
11. B,D

Do yourself - 7

1. C
2. A
3. 625 B
4. 9A
5. B
6. A
7. D
8. A
9. 2
10. 225

Do yourself - 8

1. B
2. B
3. A
4. 2
5. 1
6. 9
7. 8
8. D
9. B
10. D
11. D
12. A,C
13. B,C,D
14. A,B,D
15. $\begin{bmatrix} 17 & 4 & -19 \\ -10 & 0 & 13 \\ -21 & -3 & 25 \end{bmatrix}$
16. (a) $X = \begin{bmatrix} -3 & -3 \\ \frac{5}{2} & 2 \end{bmatrix}$, (b) $X = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$, (c) no solution

Do yourself - 9

1. $\lambda = 0, 3$ and $15 \operatorname{tr}(A) = 18, |A| = 0$ 2. A 3. B

Do yourself - 10

1. D 2. A 3. 2 4. 6 5. D

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	B	B	B	A	A	D	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	C	C	D	A	A	B	D	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	D	C	D	B	D	C	D	A	D

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B,C	A,B,C	B,C	A,B,C,D	A,B,C,D	A,B	A,B,C,D	A,C	A,D	A,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,C	A,C,D	A,B,C	A,B,C,D	B,C,D	B,D	B,C	A,B,D	A,C	B,C,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9	
Ans.	C	A	A	B	C	D	C	D	A	
Q.10	A	B	C	D						
	S	R	P	Q						

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	-3	8	10	5949	14	1	8	36	8

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	2	3	3	2	4	4	1	3
Que.	11	12	13	14	15	16	17	18	19	
Ans.	4	1	2	672	3	1	17	100	A	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C,D	C,D	A,B	C,D	B,C	B	A,B	A	1	A,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	1,3,4	2,3,4	3,4	B,C,D	5	A,B,D	A,B,C	3